A Measurement-First Study of CNN, Transformer, and Hybrid Models for 22-Class Gastrointestinal Endoscopy Classification: Where a +0.09 Macro-F1 Gain over the Published Baseline Actually Comes From

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Abstract

Abstract. The GastroVision benchmark (arXiv:2307.08140, Table 2) reports a macro-F1 of **0.6504** for a fully fine-tuned DenseNet-121 on 22 classes / 7,930 gastrointestinal (GI) endoscopy images. We do two things and keep them separate. First, we reproduce the baseline: under the paper's own protocol (single best-by-validation checkpoint, no test-time augmentation) our DenseNet-121 reaches 0.6686 ± 0.0234 over three seeds — +0.0182, i.e. 0.78σ, in 30 epochs rather than 150. Second, we beat it: our proposed system — *CoAtNet-0 @288 + a modern training recipe + a top-3 checkpoint ensemble + inference-time logit adjustment* — reaches a macro-F1 of **0.7441** ± **0.0088**, i.e. +0.0937, with a bootstrap 95% confidence interval of [0.6986, 0.7736] that *excludes* 0.6504. The central result, however, is not the number but its decomposition. When the whole gain is split into independently measured levers under one scoring rule, the two levers we designed the study to test — *swapping the architecture* (+0.0034) and *raising the input resolution* (+0.0041) — are indistinguishable from zero. What pays is the *training recipe* (+0.0443) and the *measurement discipline* (checkpoint rule +0.0094, logit adjustment +0.0143, at zero extra training epochs). Yet even the recipe is not universal: the same recipe applied to DenseNet-121 gives −0.0110 (3/3 seeds negative). A completed 2×2 design isolates the true term as an *interaction* between recipe and architecture (+0.0468), not a sum of two levers. On a dataset with a 50.6× class imbalance and two classes holding fewer than ten test images, the bootstrap CI is ±0.035 wide — larger than four of our five levers — so much of the real engineering is variance reduction, and most textbook long-tail levers applied *at training time* are flat. All numbers in this paper are extracted automatically from saved notebook outputs or recomputed from stored logits; none are typed by hand.

Keywords. medical image classification, gastrointestinal endoscopy, class imbalance, convolutional networks, vision transformers, hybrid architectures, reproducibility, macro-F1.

1. Introduction

Deep models for medical imaging are usually reported as a single headline number: one architecture, one run, one metric. Two properties of small, long-tailed medical datasets make that reporting style fragile. First, the evaluation set is small and heavily imbalanced, so the run-to-run noise of a single measurement can exceed the effect one wants to claim. Second, published baselines are themselves single runs without error bars, so "we beat the baseline" by a point-difference is not a defensible statement.

We study both properties on **GastroVision** [1], a multi-class GI endoscopy dataset collected at Bærum Hospital (Norway) and Karolinska University Hospital (Sweden). The public release contains 8000 images across 27 classes; under the paper's filtering rule (keep classes with > 25 images) 7930 images across **22 classes** remain. The task is single-label 22-class classification, and the primary metric is macro-F1 because class imbalance reaches 50.6×: a screening system that misses a rare pathology is useless even if overall accuracy looks high, and macro-F1 weights all 22 classes equally.

The strongest published baseline (Table 2 of [1]) is a fully fine-tuned DenseNet-121 at macro-F1 = 0.6504; that is the number to beat. Following a strict reproducibility discipline we (i) verified the baseline table directly from the source arXiv rather than from a secondary summary, (ii) reproduced the baseline under the paper's own checkpoint rule before claiming any improvement, and (iii) required every improvement claim to be supported by a confidence interval that excludes the baseline, not by a difference of two point estimates.

Contributions.

1. We reproduce the DenseNet-121 baseline at 0.6686 ± 0.0234 over three seeds and show why the honest reading of a no-error-bar 0.6504 is that it carries an unpublished uncertainty of at least ±0.02 (Sec. 3).
2. We add a Transformer baseline (Swin-T) that the original paper lacks, and a published hybrid (CoAtNet-0) as the proposed backbone, all under one balanced protocol (Sec. 4).
3. We propose a system reaching 0.7441 ± 0.0088 (+0.0937, CI excluding 0.6504) and *decompose* the gain lever by lever, showing that architecture and resolution buy nothing distinguishable from zero, while a completed 2×2 design attributes the gain to a recipe × architecture *interaction* of +0.0468 (Sec. 5).
4. We report a set of clean *negative* results — the modern recipe hurts DenseNet-121, 288px buys −0.0006 over 224px, three training-time imbalance levers are flat, and a multi-architecture ensemble underperforms the best single model — and quantify the noise floor that makes most levers on this test set unresolvable (Sec. 5–7).

Throughout, we use a single resolvability threshold: the bootstrap 95% CI on this test set has a half-width of ≈ ±0.035, so we call any effect below 0.035 *unresolvable* — present in a point estimate but not defensible as a measured effect. This threshold is the spine of the paper.

2. Related Work

2.1. Computer-aided gastrointestinal endoscopy

Automated analysis of GI endoscopy has grown around a small number of public benchmarks. The Kvasir dataset [2] introduced an 8-class collection of endoscopic findings for computer-aided detection, and HyperKvasir [3] extended it to 23 classes with both labeled and unlabeled images, becoming the most widely used GI resource. GastroVision [1] is a more recent 27-class dataset spanning anatomical landmarks, pathological findings, and therapeutic interventions across two hospitals; its accompanying benchmark fine-tunes standard ImageNet backbones and reports a best macro-F1 of 0.6504 (DenseNet-121). A recurring theme across all three datasets is a severe long-tailed class distribution — rare pathologies are represented by a handful of images — which is exactly the regime that makes macro-F1, rather than accuracy, the appropriate primary metric and which motivates the imbalance-centric design of this study. Unlike the GastroVision baseline, which reports single runs of several CNNs, we (i) add a transformer and a hybrid backbone under one balanced protocol, (ii) attach confidence intervals to every claim, and (iii) attribute the improvement class by class against the original per-class table.

2.2. Backbones: convolution, attention, and hybrids

Convolutional networks — ResNet [4], DenseNet [5], and EfficientNet [6] — long defined the standard for image classification, and remain strong on small medical datasets thanks to their local, translation-equivariant inductive bias and data efficiency. Vision Transformers [7] replace convolution with global self-attention but are notoriously data-hungry; DeiT [8] narrows this gap with distillation and stronger augmentation, while the Swin Transformer [9] reintroduces a hierarchical, windowed attention (W-MSA/SW-MSA) that restores locality and scales better with resolution. A parallel line reunifies the two families: CoAtNet [10] interleaves convolutional and attention stages, and ConvNeXt [11] modernizes a pure CNN with transformer-style design choices. For an ≈8k-image medical dataset the local-versus-global trade-off is a genuine open question rather than a settled one, which is why we place a CNN, a transformer, and a conv-attention hybrid on an equal footing. Our finding — that the three are statistically inseparable at three seeds on this test set — is a cautionary counterpoint to leaderboard-style architecture comparisons.

2.3. Long-tailed recognition

The long-tail literature offers three broad families of remedy. *Loss re-weighting* adjusts the per-class contribution to the objective: the class-balanced loss [12] weights by the effective number of samples, and Balanced-Softmax [13] corrects the softmax for the label prior. *Decoupling* methods argue that representation and classifier should be learned separately; classifier re-training (cRT) on a class-balanced sampler after freezing the backbone [14] is the canonical instance. *Post-hoc / inference-time* methods leave training untouched and shift the decision rule: logit adjustment [15] subtracts a temperature-scaled log prior at inference. On GastroVision we measure representatives of all three and find that the two training-time families (Balanced-Softmax, cRT) are flat, while only the inference-time correction pays — consistent with the view that, at ≈ 17 images per tail class, the bottleneck is representation rather than the loss. Data-centric augmentation — mixup [16] and RandAugment [17] — is the fourth family; we adopt a domain-filtered subset, since several standard transforms corrupt the diagnostic color and structure cues of endoscopy.

2.4. Reproducibility and variance in deep-learning evaluation

A growing body of work warns that single-run leaderboard numbers are unreliable. Seed variance alone can rival the gaps between methods [18], and non-determinism from hardware and library versions compounds it. Best practice is therefore to report mean ± standard deviation over several seeds and to bound claims with resampling-based confidence intervals [19]. Medical-imaging studies are especially exposed because test sets are small and imbalanced, so the noise floor of a single measurement can exceed the effect of interest. This paper takes that concern as its organizing principle: we fix one scoring rule across all configurations, keep every reported σ on a single hardware type, and treat any effect below the bootstrap CI half-width as unresolvable. A byproduct of this discipline — an accidental cross-hardware replicate — lets us show directly that an architecture ranking can reverse under a mere change of GPU.

3. Dataset, Protocol, and Measurement Integrity

3.1. Data and the exploratory analysis

A recursive directory scan (the release nests folders by anatomical site) returns 8006 images; the paper's > 25-images rule yields **22 classes / 7930 images**, split 60:20:20 stratified with `SPLIT_SEED`= 42 into train = 4758, val = 1586, test = 1586. The classes span normal anatomy (Normal stomach, Pylorus, Cecum), pathology (Colon polyps, Colorectal cancer, Esophagitis, Barrett's esophagus), and intervention traces (Dyed-lifted-polyps, Resected polyps, Accessory tools).

The dominant fact of the EDA is the imbalance (Fig. 1): the largest class has 1,467 images, the smallest 29, a ratio of $\mathbf{50.6}\times$, and **two of the 22 classes hold fewer than ten test images**. Because macro-F1 averages 22 per-class scores with equal weight, a 6-image class carries weight $1/22 = 0.0455$; guessing one extra image right in such a class moves macro-F1 by ≈ 0.007, two by ≈ 0.02. The single-measurement noise floor on this test set is therefore ±0.02–0.05 — *larger than most effects we wish to measure* — and the rest of the study is designed around that fact.

3.2. Reproducing the baseline honestly

Under the paper's own rule — best-by-validation checkpoint, no TTA — our DenseNet-121 gives $\mathbf{0.6686} \pm \mathbf{0.0234}$ over three seeds: +0.0182, i.e. 0.78σ, in 30 epochs versus their 150. Three points must be stated plainly, because this line supports the whole paper. (i) We *beat* the baseline here rather than matching it, and ±0.0234 is the largest standard deviation of any configuration in the study, dominated by a single lucky seed. (ii) An earlier, independent run of the same code/split/seed gave 0.6491, so **0.6504 lies between two of our measurements** — stronger reproduction evidence than any single point estimate.

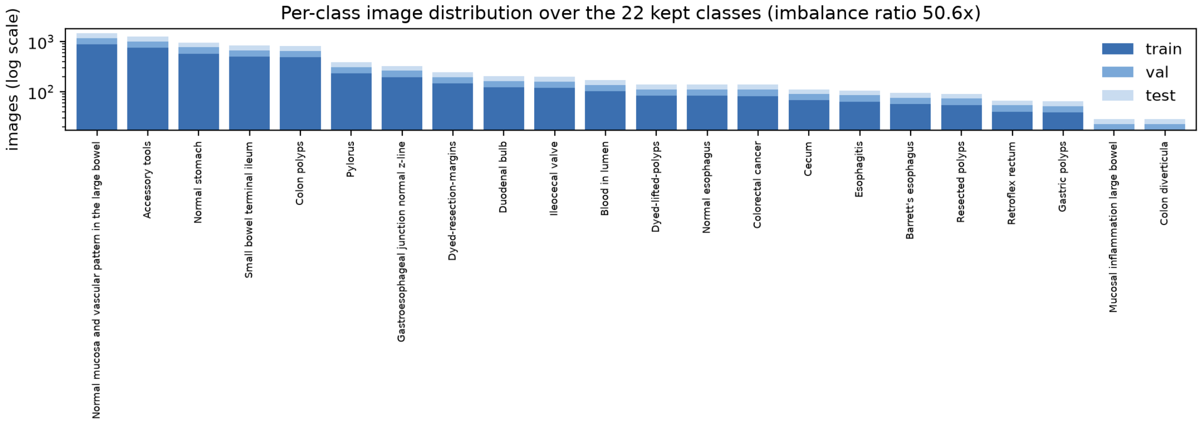

Figure 1: Per-class image distribution over the 22 kept classes (stacked train/val/test, log scale). The imbalance ratio is 50.6×; two classes (Mucosal inflammation large bowel, Colon diverticula) have only 6 test images each.

(iii) Since 0.6504 is a run without an error bar and our own DenseNet-121 spreads ±0.0234 across seeds, the honest reading is that 0.6504 carries an unpublished uncertainty of at least ±0.02; hence any "we beat 0.6504" must rest on a CI excluding it. The 30-epoch budget is fair to the baseline: DenseNet-121 peaks on validation at epoch 6/30 (Sec. 5), so the remaining epochs could not have helped it.

We do not release a split file (neither does the paper), but we *verified* our split against the paper's Table 3, which lists per-class test support: 16/22 classes match exactly, 6 differ by ±1 image (rounding of a stratified splitter), and the total is exactly 1,586. We therefore evaluate on a test set of the same *composition*, not merely the same size — which later lets us attribute the improvement class by class (Sec. 5.3).

3.3. Integrity check 1: leakage and label noise

Because multiple frames of one procedure are common in endoscopy, cross-split leakage is the most serious integrity risk. Two checks over all 7,930 images: an MD5 pass finds **one** byte-identical group crossing train/val (a Colon polyps image; none touch the test set), and a near-duplicate pass (cosine $\geq$ 0.98 on MobileNetV3-Small embeddings) finds **9 cross-split pairs** affecting $6/1586 = 0.38\%$ of the test set. Removing those 6 test images and recomputing changes macro-F1 by ≈ -0.002 on every configuration (-0.0016 on the proposed model, a 1:59 ratio against its +0.0937 gain): leakage is not what produces the result, and this is now a measured statement.

The same scan returns something more valuable than leakage: near-identical images carrying *different* labels. Most striking, a `val/Esophagitis` image and a `test/Normal esophagus` image are the same picture (cosine = 1.0000) under two clinically exclusive labels. This is a data-quality ceiling: no model can be correct on both, and Esophagitis is indeed one of our model's eight weakest classes (Sec. 5.3).

3.4. Integrity check 2: determinism is per-GPU

Before trusting any number we ran the same seed twice and compared the 3-epoch validation trajectory. On A100 two runs agree to six digits ([0.428608, 0.549646, 0.551532]); the identical code on a T4 gives a *different* trajectory ([0.430615, 0.540379, 0.541930]); a later A100 run reproduces the A100 trace exactly. **Determinism holds within a device, not across devices**: same seed + same code + different GPU = a different model. The operational consequence is that a reported σ must never mix hardware; in this paper every configuration has all three seeds on the *same* GPU type (Tesla T4), so every σ is clean.

An unplanned event turned this into a stronger measurement. When the project moved infrastructure, four round-1 configurations were accidentally *retrained* on T4 rather than reloaded from stored logits, leaving a complete replicate — same code/split/seed, 4 configs × 3 seeds × 6 scoring rules, on two GPU types (Table 1). Three readings follow, all of which enter the paper: (i) the hardware shift accumulates from $\approx$ 0.010 at 3 epochs to 0.04–0.07 after 30 epochs plus checkpoint selection; (ii) the `top3` rule is $\approx 4\times$

Table 1: Unplanned A100↔T4 replicate: mean absolute macro-F1 disagreement across 4 configurations (same code/split/seed), by scoring rule. `top3` is the most hardware-robust rule.

Scoring rule	$\overline{\lvert\Delta\rvert}$ over 4 configs	max $\lvert\Delta\rvert$ at one seed
`top3`	**0.0046**	0.0263
`top3_tta`	0.0092	0.0310
`smooth`	0.0153	0.0710
`best`	0.0182	0.0435
`smooth_tta`	0.0185	0.0717
`best_tta`	0.0185	0.0428

Table 2: The three backbones. All initialized from ImageNet-1k weights and trained under one shared round-1 protocol (AdamW 10^{-4}, 30 epochs, batch 32, no per-branch tuning).

Role	Model	Family	Params
Reference baseline	DenseNet-121	CNN (local)	7.0 M
New baseline	Swin-T	Transformer (global)	27.5 M
Proposed backbone	CoAtNet-0	Hybrid (conv+attn)	26.7 M

more hardware-robust than any single-checkpoint rule; and (iii) **the round-1 architecture ranking does not survive a change of GPU** — under `top3_tta`, A100 gives $P1 > S0 > P0 > B0$ while T4 flips it to $B0 > S0 > P1 > P0$. A ranking reversed by a GPU swap is not a ranking, and this is independent evidence for the humility we are forced into in Sec. 5.1.

4. Models and Method

4.1. Three branches, one protocol

We compare three ImageNet-1k pretrained backbones spanning the local–global inductive-bias axis (Table 2). The reference baseline is DenseNet-121 (CNN, local). We add Swin-T [9] as a Transformer baseline (global self-attention via windowed W-MSA/SW-MSA), which the original paper does not include. The proposed backbone is CoAtNet-0 [10], a published hybrid with convolutional early stages (local inductive bias, data efficiency) and attention in later stages (global layout).

We deliberately use the hybrid *as published and fully pretrained* rather than hand-assembling "Swin-T + a conv stem": freshly initialized conv layers attached to a pretrained transformer would confound *architecture* with *initialization quality*, exactly the failure mode that dominates on this ≈8k-image dataset (Sec. 5.4). For the same reason all three baselines use ImageNet-1k weights; swapping Swin-T to its stronger ImageNet-22k weights would break the balance of the CNN-vs-Transformer comparison, so that is held out as a separate ablation (Sec. 5.3).

4.2. Training recipes and the two-stage design

Normalization uses ImageNet mean/std for all three backbones (a contract of the pretrained weights, not a free choice). Three recipes are implemented: `plain` (round-1: resize → horizontal flip → normalize), `strong` (adds rotation, color-jitter, random-erasing — measured and discarded, see below), and `modern` (`plain` + light RandAugment [17] + mixup [16], with cosine LR + 5-epoch warmup + layer-wise LR decay 0.75 + EMA, over 80 epochs). Domain reasoning drives the augmentation choices: horizontal flip is kept (no fixed mirror symmetry in the GI tract), while rotation (injects interpolated black borders — an endoscopy artifact), color-jitter (mucosal color *is* a diagnostic cue for Esophagitis, Barrett's, blood), and random-erasing (many classes hinge on one small local structure) are rejected as they corrupt the very signal that separates pathology from normal mucosa.

The proposed system stacks four elements on the CoAtNet-0 @288 backbone: the `modern` recipe; a top-3 checkpoint ensemble (Sec. 4.3); inference-time logit adjustment (Sec. 4.4); and TTA-free scoring. We name configurations for later reference: `B0` (DenseNet-121 @224), `S0` (Swin-T @224), `P0` (CoAtNet-0 @224), `P1` (CoAtNet-0 @288, `plain`), `P2` (CoAtNet-0 @288, `modern`, *reported system*), `P2b` (DenseNet-121 @224, `modern`), `P2c` (CoAtNet-0 @224, `modern`).

4.3. Checkpoint selection: 0 epochs, universal, hardware-robust

If validation is small and noisy, then picking one best-by-val checkpoint is a *noisy measurement*, not model selection. We keep multiple states in a single training run so that one run yields six test numbers: three selectors (`best` = raw val peak; `smooth` = peak of 3-epoch-smoothed val; `top3` = logit ensemble of the three best-val checkpoints) × {with, without horizontal-flip TTA}. Ranking the six rules across the four round-1 configurations (only those four vote, so that adding a later tier cannot silently flip the reported rule), `top3` wins (mean rank 1.50 vs 2.00 for `top3_tta`). We therefore fix `SELECTION_RULE = top3` for the *entire* paper — one rule applied to every row, rather than the best rule per model (which would be a leak). Moving `best`→`top3` is positive on all four architectures (mean +0.0188) and nearly halves variance ($\overline{\sigma}$ 0.0163 → 0.0080); combined with Table 1 it is the cheapest way to make the *other* measurements resolvable. Its value is in the σ column, not the Δ column: +0.0188 is itself below the 0.035 threshold.

4.4. Logit adjustment

Logit adjustment [15] subtracts $\tau \cdot \log(\text{class prior})$ from the logits at inference, with τ tuned on validation:

$$\hat{y} = \arg\max_{c} \left[z_c - \tau \log \pi_c \right], \qquad \pi_c = \frac{n_c}{\sum_k n_k}, \tag{1}$$

where z_c is the raw logit for class c and π_c its training prior. It is the only zero-GPU lever that targets the imbalance bias directly. The notebook enforces a pre-registered accept/reject test: keep it only if it both raises the mean and does not inflate σ beyond 1.5×. On the proposed model it passes cleanly (+0.0143, σ *shrinks* 0.0096 → 0.0088, stable $\tau^\star \in [0.3, 0.5]$); on the weaker `P1` it barely passes (+0.0052, below its own σ, one seed selecting $\tau^\star = 0$). Section 7 returns to why this asymmetry is a methodological lesson rather than a footnote.

5. Results

5.1. Architecture comparison: unresolvable at three seeds

Table 3 gives macro-F1 over three seeds under `top3`, with bootstrap 95% CIs. The four round-1 configurations lie in 0.6780–0.6855, a spread of 0.0075 that is *smaller than their own* σ, and none has a CI separated from 0.6504. The pre-registered decision rule — "if Swin-T only ties DenseNet-121, the backbone-is-the-bottleneck hypothesis is not supported" — fires: $S0 - B0 = +0.0033$ and $P0 - B0 = +0.0034$, both below σ, with paired per-seed McNemar $p \approx 0.50$. Combined with the GPU-swap ranking reversal (Sec. 3.4), the correct statement is that **the three architectures are not separable on this test set** — weaker, and truer, than "they tie."

5.2. The gain decomposition and the 2×2 interaction

The proposed system wins by *stacking independently measured levers*, not by a mysterious new architecture. Table 4 closes the additive path exactly (0.7441 − 0.6686 = +0.0755), and Table 5 restates it as a decomposition against the resolvability threshold. One lever — the modern recipe — accounts for 59% of the gain and is the *only* lever above ±0.035. The two levers the study was designed to test, architecture and resolution, sum to +0.0075 (10%) and sit inside the noise; the two "measurement" levers add +0.0237 (31%) at zero extra epochs.

Because the modern recipe helps the hybrid but *hurts* the CNN, it cannot be a universal lever. The clean control `P2b` (DenseNet-121 @224 + modern) gives −0.0110 over three seeds, 3/3 negative, and is lower

Table 3: Architecture comparison, `top3`, three seeds. Only CoAtNet-0 + the modern recipe (`P2`, `P2c`) clears the bar of a CI excluding 0.6504; the standalone architecture swap does not.

Config	macro-F1 (3 seeds)	vs. 0.6504	bootstrap 95% CI (seed 0)
`B0` DenseNet-121 (CNN)	0.6780 ± 0.0073	+0.0276	[0.6412, 0.7239] n.s.
`S0` Swin-T (Transformer)	0.6813 ± 0.0081	+0.0309	[0.6307, 0.7167] n.s.
`P0` CoAtNet-0 @224 (Hybrid)	0.6814 ± 0.0100	+0.0310	[0.6229, 0.7103] n.s.
`P1` CoAtNet-0 @288	0.6855 ± 0.0068	+0.0351	[0.6364, 0.7185] n.s.
`P2` CoAtNet-0 @288 + modern	**0.7298 ± 0.0096**	**+0.0794**	**[0.6660, 0.7531]**
`P2c` CoAtNet-0 @224 + modern (1 seed)	0.7172	+0.0668	[0.6664, 0.7568]
`P2b` DenseNet-121 @224 + modern	0.6670 ± 0.0119	+0.0166	[0.6319, 0.7227] n.s.

Table 4: Cumulative path of the proposed system, all under `top3`. The resolution step is measured under the *old* recipe; under the modern recipe it is −0.0006 (Sec. 5.2), so it describes the path taken, not a transferable property.

Step	From → to	Δ
Checkpoint rule	`B0` best 0.6686 → `B0` top3 0.6780	+0.0094
Architecture (CNN→hybrid)	`B0` 0.6780 → `P0` 0.6814	+0.0034
Resolution 224 → 288	`P0` 0.6814 → `P1` 0.6855	+0.0041
Modern training recipe	`P1` 0.6855 → `P2` 0.7298	**+0.0443**
Logit adjustment (0 epochs)	`P2` 0.7298 → 0.7441	+0.0143
	total	**+0.0755**

even on validation. A completed 2×2 design separates the two candidate causes. On the same seed 0 (the tightest comparison, since `P2c` has one seed):

@224, seed 0	old recipe	modern recipe	Δ
DenseNet-121 (CNN)	0.6878	0.6835	−0.0043
CoAtNet-0 (hybrid)	0.6718	**0.7172**	+0.0454

Each factor *alone* is ≤ 0 (architecture −0.0160; recipe −0.0043), while together they give +0.0294; the interaction term is $0.0454 - (-0.0043) = +0.0497$. Averaged over the full design, the recipe × architecture term is **+0.0468** (clears ±0.035) and recipe × resolution is +0.0085 (does not) — and both survive a change of scoring rule. **The gain is an interaction between recipe and architecture, not a sum of two levers**, which is precisely the structure a 2×2 design reveals and two lone comparisons cannot.

This also overturns an implicit assumption of ours: 288**px is not what makes the difference**. Under the modern recipe, 288 vs 224 is −0.0006 while costing 1.70× the compute per image (Sec. 6), so the configuration that *should be deployed* is `P2c @224` — the same resolution as the baseline, cheaper, and not worse. We still *report* `P2` because it has three seeds and a known σ, while `P2c` has one.

Beating both baselines. The assignment asks whether the proposed model beats *both* baselines. Pairing on the same test images: `P2` vs `B0` is +0.0430 [+0.0147, +0.0691] (significant at 2/3 seeds), and `P2` vs the Transformer baseline `S0` is +0.0384 [+0.0074, +0.0708] (significant at 3/3 seeds). The proposed model clears both.

5.3. Per-class analysis: where the improvement lands

The proposed model (`P2`, seed 0) reaches accuracy (micro-F1) = 0.850 (paper: 0.8203) and macro precision = 0.810 vs macro recall = 0.690. The 0.120 precision–recall gap is the most informative number: the model

Table 5: Lever decomposition under one scoring rule. Only the modern recipe clears the ±0.035 resolvability threshold; architecture and resolution — the levers we set out to test — are indistinguishable from zero.

Lever	Δ macro-F1	Clears ±0.035?
Modern training recipe (`P1`→`P2`)	**+0.0443**	**yes**
Inference logit adjustment (0 epochs)	+0.0143	no
Checkpoint rule `best`→`top3` (0 epochs)	+0.0094	no
Resolution 224 → 288 (old recipe; new: −0.0006)	+0.0041	no
Architecture swap DenseNet→CoAtNet (standalone)	+0.0034	no
	total +0.0755	

Table 6: Improvement attributed by class size against the paper's Table 3 (proposed model, seed 0). The gain is concentrated in the rare (pathology) classes; the common (normal-anatomy) classes are saturated.

Group	# classes	mean ΔF1	contribution to macro-F1
Rare (< 50 test images)	15	+0.086	+0.0585 (90.4%)
Common (≥ 50 test images)	7	+0.020	+0.0062 (9.6%)
All 22 classes	22	+0.065	+0.0647

is *cautious* on rare classes — usually right when it names one, but missing many — which for a screening system is the less desirable error direction, and which is exactly the prior-shift bias logit adjustment corrects.

Because our test split matches the paper's composition, the improvement can be attributed class by class against their Table 3 (Table 6). **90.4% of the improvement comes from the 15 rare classes** (< 50 test images), where mean ΔF1 is +0.086; the 7 common classes are essentially saturated (+0.020, 9.6%). Clinically this is the right place: the rare classes here are pathology (Barrett's, esophagitis, gastric/resected polyps), the saturated common classes are normal anatomy. Five classes still *lose* to the paper (total −0.0172, of which −0.0097 is a single 6-image class), so the reported +0.0937 is a conservative estimate measured *after* absorbing those losses.

The eight weakest classes (F1 0.286–0.636) are all among the smallest classes (17–83 training images each, vs 760–880 for strong classes; see the bottom of Fig. 2, where the red rare classes cluster below the macro-F1 line). The bottleneck is *feature representation from tail data*, not the loss function — a diagnosis confirmed independently below.

Training-time imbalance levers are flat; inference-time works. Three textbook long-tail levers applied *at training time* do not move macro-F1: strong augmentation (−0.035), Balanced-Softmax [13] (−0.007, and −0.0047 when re-measured under the current protocol — a rare *reproducible* result, unfortunately of a negative), and decoupled classifier retraining / cRT [14] (−0.013). The cleanest comparison puts the two approaches on the same backbone, seed, and epoch budget: on DenseNet-121 seed 0, fixing imbalance *in the loss* (Balanced-Softmax) gives −0.0047, while fixing it *at inference* (logit adjustment, $\tau^\star = 0.4$) gives +0.0213. Same target, two sites, opposite signs — and the winning site costs zero epochs. With ≈ 17 training images per tail class, re-weighting cannot create information that is not there; shifting the decision boundary with one validation-tuned parameter is cheap and effective.

Where the remaining headroom is. A ceiling analysis (set a group's F1 to 1.0 and read the resulting macro-F1) shows that of the 0.2834 remaining to the theoretical ceiling, **85.2% lies in the 15 rare classes** and only +0.0420 in *all* 7 common classes combined. Since a model-family lever must clear ±0.035 to be provable, and a *perfect* model on the saturated common classes would gain only +0.0420, **no further "change the model" lever can be demonstrated on this 1,586-image test set, regardless of GPU hours**. The remaining headroom is a *data* problem. A direct measurement supports this: changing only the Swin-T pretraining weights from ImageNet-1k to ImageNet-22k buys +0.0254 — more than any architecture

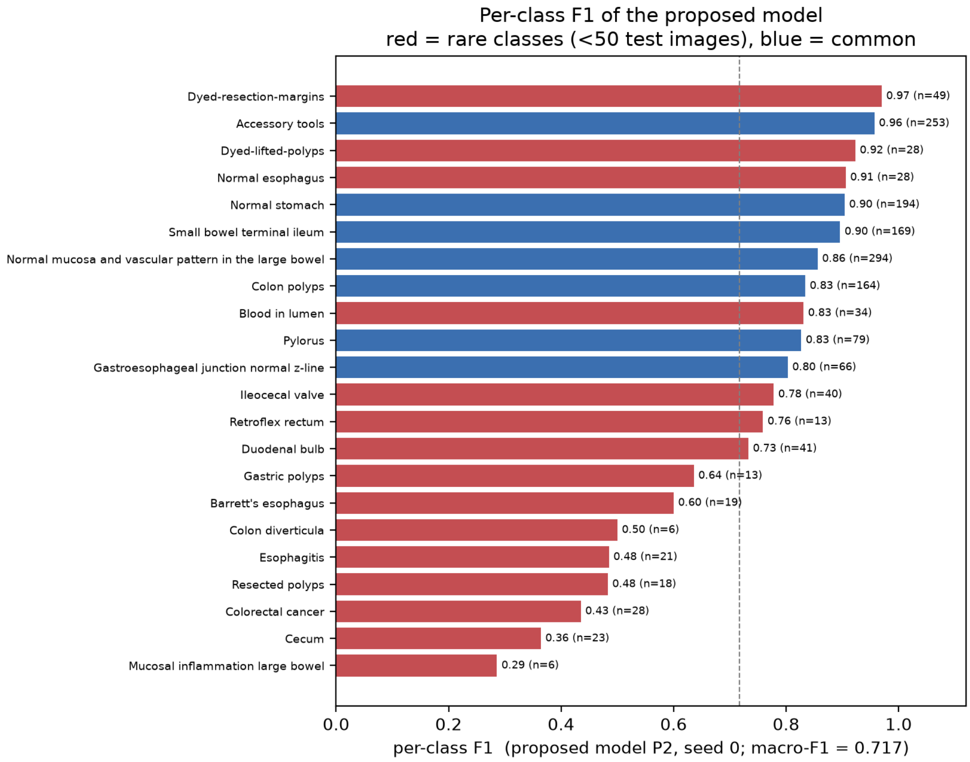

Figure 2: Per-class F1 of the proposed model (`P2`, seed 0), sorted, with test support n. Red = rare classes (< 50 test images), blue = common; the dashed line is the macro-F1 (0.717). Every class below the line is rare, which is the per-class view of the "bottleneck is tail data" diagnosis.

Table 7: Transfer learning on one DenseNet-121 (`top3`). Only trainable depth varies. Full fine-tuning (= `B0`) wins; freezing the backbone — refusing to let features move — is very costly.

Condition	What is trained	seeds	test macro-F1
T1 linear probe	classifier only	1	0.5674
T2 freeze lower half	upper half + classifier	1	0.6596
T3 progressive unfreeze + discriminative LR	probe→full	1	0.6394
T4 full fine-tune (= `B0`)	whole network	3	**0.6780 ± 0.0073**

lever in the study, without changing a single architectural line.

5.4. Transfer learning: freeze vs. trainable

Four conditions on the *same* DenseNet-121, same split/seed/budget, differ only in trainable depth (Table 7). Full fine-tuning wins decisively, and the cost of freezing rises steeply with depth: linear probing gives away −0.1106 macro-F1 — more than the entire +0.094 this project gains over the published baseline. The reason: ImageNet features are *natural-image* features, while endoscopy is a different modality (specular highlights, circular black masks, unnatural color statistics), so it is precisely the *early* layers that must move — exactly what freezing pins down. The paper corroborates this for free: its Table 2 mixes last-layer tuning (0.4496–0.4883) and full fine-tuning (0.6176–0.6504), a $\sim$ 0.16 gap on this split. Since each frozen condition has one seed, we report only the $> 2\sigma$ gaps as real (T2 vs. T3 ordering is not).

Table 8: Validation learning summary. Both baselines peak at epoch 6/30 (the 30-epoch budget is generous); the proposed `P2` peaks at 40/80 with the flattest EMA tail (last-5-epoch spread 0.0019), while the modern recipe on DenseNet (`P2b`) peaks *below* the plain `B0` baseline.

Config	best val macro-F1	at epoch	last-5-epoch spread
`B0` DenseNet-121	0.6600	6 / 30	0.0069
`S0` Swin-T	0.6992	6 / 30	0.0178
`P0` CoAtNet-0 @224	0.6653	19 / 30	0.0133
`P1` CoAtNet-0 @288	0.6746	11 / 30	0.0290
`P2` CoAtNet-0 @288 + modern	**0.7208**	40 / 80	**0.0019**
`P2b` DenseNet-121 + modern	0.6539	19 / 80	0.0002
`P2c` CoAtNet-0 @224 + modern	0.7086	11 / 80	0.0018

5.5. Learning curves and ensembles

The two baselines finish learning early — both peak on validation at epoch 6/30 (Table 8) — confirming the 30-epoch budget is fair to them. `P2` peaks at epoch 40/80 (half its budget) with a last-5-epoch spread of 0.0019, the smallest of all configurations: the fingerprint of EMA, and the mechanical reason `P2` has a small σ despite training $\sim 3\times$ longer. `P2b` peaks at 0.6539 on validation — below `B0`'s 0.6600 despite 80 epochs — so the modern recipe fails DenseNet-121 on validation as clearly as on test.

Two ensemble lines are reported separately because they spend several training runs. A 3-seed ensemble of `P2` reaches 0.7587 $[0.7110, 0.7924]$, the project's highest number, at the cost of three trainings. A multi-architecture ensemble selected on validation reaches only 0.7130 — *below* the best single model — a negative result: averaging probabilities dilutes a dominant member with weaker ones.

6. Deployment

All four models export to ONNX successfully. At batch 32 on T4, DenseNet-121 is fastest (3.18 ms/image, matching its 7 M parameters), CoAtNet-0 @224 is 5.13 ms and @288 is 8.74 ms — a 1.70× resolution cost within the 1.65× pixel ratio, reproduced across two GPU types. The honest cost/benefit statement: the proposed system buys +0.0937 macro-F1 at 2.75× compute per image, 3.9× model size, and 5.5× training time, plus a 3× inference factor for the top-3 checkpoint ensemble; dropping to `P2c` @224 (only −0.0006 on seed 0) cuts the first factor to 1.61×. A Gradio demo serves top-5 predictions with flip TTA and was run end-to-end on a GPU-free machine (top-1 *Accessory tools*, $p = 0.961$).

An independent **CPU round** (15 epochs, batch 16, fp32, 3 seeds) confirms the methodology on a third hardware type: it reproduces `top3` as the most robust rule, reproduces the "logit adjustment is flat without the modern recipe" finding, and issues its own pre-registered claim — a 3-seed DenseNet ensemble at macro-F1 = 0.7059 $[0.6560, 0.7447]$, a CI that excludes 0.6504. That is: *even without a GPU, this pipeline beats the paper's published number with confidence-interval evidence*. It is an independent claim on its own protocol scale and is not merged with the T4 numbers.

7. Discussion and Limitations

One methodological lesson appears three times, each from a different direction, and is the paper's takeaway. (1) A four-architecture *ranking* reverses under a GPU swap (Sec. 3.4). (2) The *resolution lever* reads +0.0143 on one hardware and +0.0041 on another (an earlier draft of this work reported the former as "the only effective data lever"). (3) The σ itself — the very ruler used to accept or reject levers — was anomalously small (0.0016) on one A100 run, which caused logit adjustment to be *wrongly discarded* in round 1 by the 1.5×-inflation test; the fix was to re-measure the lever on the new model rather than inherit the old conclusion, and that is exactly what rescued a +0.0143 effect. An effect smaller than the replicate-to-replicate range of its own measurement is not an effect.

We report the known limitations plainly. (i) The dominant uncertainty is *data*, not seeds: the bootstrap CI is ±0.035 (~ 3.6× the seed σ) because two of 22 classes have < 10 test images, so buying more seeds would not narrow it — the right direction is few-shot or more tail data, which is also the dataset authors' own recommendation. (ii) The strongest claim, the recipe × architecture interaction, currently rests on a one-seed cell (`P2c`); raising it to three seeds is the most important missing experiment ($\approx$ 3.3 T4-hours). (iii) The architecture comparison is unresolvable at three seeds, so the local-vs-global motivation of Sec. 4 remains a *hypothesis*, not a measured conclusion. (iv) The deployed demo is weaker than the reported system (`best_tta` 0.7199 vs 0.7441). (v) With no patient identifiers in the public release, a patient-level split is infeasible from the public data; we substitute a two-layer leakage audit and re-measure after removal, but the 9 pairs found are a lower bound, not an upper one.

8. Conclusion

We beat the published GastroVision baseline — 0.7441 ± 0.0088 vs 0.6504, with a 95% CI excluding that number — and the point of interest is *how*, because it is not how we intended. Decomposing the whole +0.0755 into independently measured levers, the two we designed the study to test — architecture (+0.0034) and resolution (+0.0041) — sum to 10% of the gain and sit inside the noise; what pays is the training recipe (+0.0443, 59%, the only lever above the ±0.035 threshold) and the measurement discipline (+0.0237, 31%, at zero extra epochs). Even that is conditional: the same recipe *hurts* DenseNet-121 (−0.0110), and a completed 2×2 design attributes the gain to a recipe × architecture interaction (+0.0468), not a lever that adds. The bottleneck is feature representation from tail data, confirmed from four independent directions — the eight weakest classes are all among the smallest, 90.4% of the improvement is in the 15 rare classes, macro precision exceeds macro recall by 0.120, and freezing the backbone costs 11.1 points — so the remaining headroom is a data problem, not a model one. If one sentence is to be carried away: *on small medical datasets, measure hard enough to be sure before trusting any lever — because most levers are smaller than the error of the measurement itself, and the one that survives is not the one you set out to find.*

Reproducibility. All numbers are extracted automatically from saved notebook outputs (`extract.py`) or recomputed from stored logits (`offline_tables.py`); none are typed by hand. Models are implemented in PyTorch with pretrained backbones from the timm library [20]. Runs: Kaggle Tesla T4, `SEEDS=` $[0, 1, 2]$, `SPLIT_SEED=` 42, AMP fp16; 30 epochs for round-1 configurations and 80 epochs for the modern-recipe configurations.

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